

Performance Analysis of Pairs Trading Strategy Utilizing High Frequency Data with an Application to KOSPI 100 Equities

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Abstract

In this study, the well-known pairs trading strategy, one of typical market neutral strategies, is modified to be able to utilize high frequency equity data, and it is applied to the constituent shares of the KOSPI (Korea Composite Stock Price Index) 100 index. This study is distinguished from the most of previous work on the traditional pairs trading strategy in that we introduce the use of high frequency data in strategy modeling instead of daily closing prices, which allows us to analyze the performance of the strategy in high frequency domain. More specifically, we extract the trading signal, which is based on the spread between stocks of pair, by estimating time adaptive regression coefficients using the Kalman filter. As for underlying universe for the strategy, we confine ourselves to consider the most liquid 100 stocks in KOSPI as a basket for our experiment. Major findings include that arbitrage profitability is in fact present without being subject to market condition even when conservative transaction costs are taken into account. In particular, our strategy outperforms better in bear market condition while it varies depending on industry group. The results also demonstrate that the performance of the strategy is dependent upon timing of the market entry; the performance of trades entered around at opening and closing of the daily market is appeared to be relatively superior to that of trades executed in the rest of daily trading hours. Furthermore, we introduce an enhanced version of the strategy, which selects high-ranking pairs to trade for the next time period based on a set of in-sample statistics. It is verified that the enhanced strategy has better profitability and reliability compared to our basic strategy.

Keywords : Pairs trading, Statistical arbitrage, Kalman filter, High frequency trading, KOSPI 100

I. Introduction

In this article, the well-known pairs trading strategy, one of typical market neutral long-short relative value strategy, is modified to be able to utilize high frequency equity data, and it is applied to the constituent shares of the Korean stock market.

1. Pairs trading strategy

The pairs trading is a strategy that tries to identify arbitrage opportunity based on historical equilibrium in spread between the share prices. Basically an investor evaluates the current position of the spread based on its historical fluctuations, and seizes the moment when the current spread deviates from its historical mean level by a pre-determined threshold. If the spread exceeds the threshold indeed, then the investor bets on the reversion of the current spread to its historical mean. Since this strategy focuses only on the spread having stationary properties to guarantee that the spread would mean-revert eventually, it can be categorized into a statistical arbitrage and convergence trading strategy.

2. Application of the high frequency data to strategy modeling

As mentioned earlier, we introduce the use of the high frequency equity data in strategy modeling, although the industry practice for market neutral hedge funds is to use daily sampling frequency of equity data in designing trading model. In this study, intraday stock prices data sampled at 30-minutes interval is used for the strategy, and the performance is analyzed in high frequency domain. The motivation for this study is based on the fact that the profitability of these strategies has deteriorated recently, See Gatev et al. (2006), and it has been shown that using the high frequency data is advantageous to achieve higher information ratio compared to the use of daily closing prices, see Aldridge (2009). Thus, we expect to obtain an edge over other traders using the already well-known strategy, by modifying it into new model utilizing intraday data. Besides, trading based on daily closing prices does involve a few practical limitations; market entry is only allowed during trading hours, and apparently there is possibility that trading signal is identified when market closes. Also it is legally prohibited to short sell shares at closing price in the Korean stock market.

3. The aims of the study

The primary aim of the study is three-fold; the first aim is to modify the traditional pairs trading model into a model capable of processing high frequency equity data to generate trading signal. The second is to analyze the performance of the strategy in various aspects to gain insights into the nature of high frequency market neutral investment strategy. The third aim is to establish an enhanced version of the strategy for better profit potential.

4. The brief overview of the study

This study is the first practice in the realm of high frequency market neutral trading strategy that extracts trading signal by estimating time adaptive regression coefficient using the Kalman filter scheme. As for underlying universe for the strategy, we confine ourselves to consider top 100 stocks having larger trading amounts in KOSPI as a basket for our experiment. This is to get rid of other external variables that may add undesirable noises to the overall performance which would make it difficult to analyze pure performance of the strategy itself. We analyze the results of out-of-sample performance test from various angles. We highlight how performance varies depending on market condition, industry group, and timing of the market entry. The results are quantified by utilizing a set of performance statistics such as average rate of return, winning ratio and the information ratio etc. Furthermore, we introduce an enhanced version of the strategy for better profit potential and reliability, which selects 5 high-ranking pairs as our trading portfolio for the next time period, based on a set of in-sample statistics: in-sample mean reversion strength, in-sample information ratio and in-sample t-stat of the ADF (Augmented Dickey-Fuller) test.

5. Organization of the paper

This rest of the paper is organized as follows. In section II, we present the literature review, section III describes the data used and explains the methodology implemented. Section IV gives the results of out-of-sample performance test for the strategy while it also presents the results of the enhanced trading portfolio. Section V concludes.

II. Literature Review

1. Market neutral strategy and high frequency data

The traditional pairs trading strategy has been widely used in the market since it was first pioneered by Gerry Bamberger and later led by Nunzio Tartaglia's quantitative group at Morgan Stanley in the 1980s, see Gatev et al (2006). It is known as a market neutral trading strategy enabling traders to profit from virtually any market conditions. Also it can be categorized as a statistical arbitrage or convergence long-short strategy. The strategy has been widely documented as well in current literature including Enders and Granger (1998), Vidyamurthy (2004), Dunis and Ho (2005), Lin et al. (2006) and Khandani and Lo (2007). Meanwhile, also there has been a wide range of studies of the nature of high frequency financial data including Ghysels et al. (2006) and Faust et al. (2007). However, literature examining statistical arbitrage strategy utilizing high frequency financial data is extremely limited. It is highly likely that practitioners, especially ones in hedge funds industry, have extensively developed and implemented those strategies for their businesses, propelled by major advances in computing power and cutting edge trading infrastructure. But obviously there have been few studies and practices published or reported to either academia or industry. From an extensive review of literature, there appears to be only one relevant study regarding high frequency market neutral trading strategies, see Nath (2003), which looks at market neutral strategies in US fixed-income market. In Korea, there just have been a few studies about the traditional pairs trading strategy by Boguen Choi (2006), Wonhyuk Yoo (2007), and Kwangchul Son (2010). Research on trading strategies using high frequency data has not been published or reported yet.

The fundamental principles of pairs trading strategy are conceptually simple, but there are several things to be examined carefully when implementing the strategy. We present the approaches chosen for our study with literatures reviewed as follows.

2. Time adaptive model – the Kalman Filter

We introduce a novel approach to generate true trading signals in this study. Real time high frequency data is directed to pass through the Kalman filter, producing

time-varying regression coefficients. Then with this coefficients, spread between stocks comprising pair is computed and standard normalized. The Kalman filter is a forward looking methodology, as it tries to predict the future position of the time varying regression coefficient. Although there are other approaches for adaptive parameter estimation such as rolling OLS (Ordinary least squares) regression, and DESP (Double exponential smoothing-based prediction) models, see Bentz (2003) and LaViola (2003), it has been shown that the Kalman filter method is considered as a superior technique for estimating adaptive parameters, especially when dealing with data having higher volatility, see Dunis and Shannon (2005), Choudhry and Wu (2009), Gomez (2009) and Burgess (1999).

3. Cointegration

The standard normalized spread is then the ADF (Augmented Dickey-Fuller) tested to see if corresponding stocks have cointegration relationship. Basically, there are two approaches to choose matching pair; one is the factor-based approach and the other is the cointegration-based approach. In the factor-based approach, major factors having a considerable impact on the return for each pair are defined and listed up first. Then the expected return for each pair is measured based on the factors found, and then based on the expected returns potential matching pairs are selected. Meanwhile, the cointegration-based approach is a quantitative technique that tries to find long-term relations between asset prices by only referring to stock prices data, introduced in a seminal paper by Engle and Granger (1987). If one stock is cointegrated with another stock in given time period, which means the spread between the stock prices is bounded around equilibrium level in the period, not wandering off to infinity. More advanced cointegration approach was developed by Johansen (1988), which can be applied to more than two assets at the same time. It became a well known quantitative technique widely used even in empirical research in verifying relations between multiple variables coming from a variety of academic areas. There have also been studies of market neutral arbitrage strategies based on the cointegration approach, including Burgess (2003), Lin et al. (2006), and Galenko et al. (2007). In this paper, as we only deal with pairs of shares, we prefer the simpler Engle and Granger methodology, which is implemented by using the ADF test scheme.

4. Ornstein-Uhlenbeck process

If the standard normalized spread series get through the first screening filter, the ADF filter without being flagged as invalid, then it is formulated into the OU (Ornstein-Uhlenbeck) process, producing estimated mean-reversion strength which evaluates the final availability of the spread to generate trading signal, see Mudchanatongsuk et al. (2008) for more detailed discussion of the process. This scheme is to be illustrated in more detail later in the following subsection.

III. Data and Methodology

1. Data

The data used in this study is collected from the database maintained by FnGuide, a commercial financial data provider operating in Korea. The data set covers the horizon from the 1st of October 2008 to the 31st of July 2010, which includes bullish, bearish, and flat market period within the horizon. Besides, by including the period of global financial crisis in the second half of the year 2008 within our horizon, we try to highlight the reliability of our strategy during the time when stock market melt down and global financial systems collapsed. Meanwhile, in regard to how in-sample estimation period and out-of-sample evaluation period is divided, it is worth noting that moving window with the size of one week i.e. 5 trading days is assumed to be the out-of-sample period and the period of the previous two weeks is assumed to be the in-sample period. Thus, the in-sample period and the out-of-sample period are updated and assigned new data accordingly at every end of the week. Constructing the data set in this manner ensures that the performance of the strategy generated in back-testing is similar to that realized in real trading. The construct of the in-sample and out-of-sample period is illustrated in Figure 1.

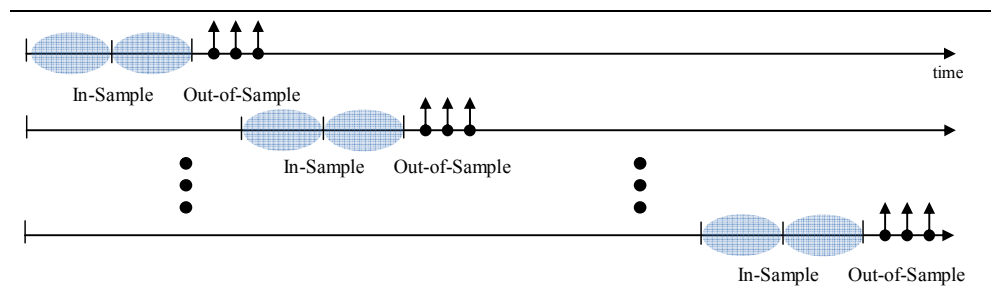


Figure 1. An illustration of the construct of the in-sample and out-of-sample period

Intraday stock prices data sampled at 30-minutes interval is used for modeling the strategy. We call this dataset as high frequency data for brevity purpose. As mentioned earlier, the aim of using high frequency data is to avoid a few practical limitations that a strategy based on daily closing prices does involve, and to secure higher potentially achievable information ratio. As for underlying universe for the strategy, we confine ourselves to consider the most liquid 100 stocks.

2. Methodology

2.1. Pairs selection

Among 100 stocks comprising KOSPI 100 index, stocks having average trading amounts in the past 60 days less than 2 million dollars¹ are discarded to lower the liquidity risk. And among all the possible pairs constructed from the remaining stocks, pairs in which each stock belongs to different industry group² are discarded to adjust the each share's exposure to the industry risk to a similar level, thereby to lower the market beta risk. Lastly among the remaining pairs, pairs having beta³ spread, calculated based on CAPM (Capital Asset Pricing Model), larger than 0.2 are discarded to realize the market neutral strategy. We are left with 65 pairs at the final stage of the selection process in our study.

2.2. Trading signal generation

Once we obtain a group of pairs to focus on, we apply the Kalman filter scheme to high frequency prices data to produce the estimate of the time-varying regression coefficients for each pair. Then spread between stocks is computed for each pair with the coefficients obtained.

The Kalman filter is a recursive algorithm which allows one to upgrade model estimates using new information. Basically the Kalman filter is based on the representation of the dynamic system with a state space regression model. It

¹ The amount of 2 million dollars is determined based on the residual quantity per bid price and the level of liquidity observed during daily trading hours.

² In this study, FIGCS (FnGuide Industry Group Classification Standard) is used as a standard to classify stocks, which lists up 25 industry groups in it.

³ For CAPM, the return of the market index is calculated from KOSPI 200 index series for corresponding timeframe.

formulates the beta dynamics through an autoregressive process driven by a white noise process. In our study, the state-space representation of the dynamics of the time-varying regression coefficients is given by the following system of equations:

$$P_{y_t} = P_{x_t} \beta_t + \varepsilon_t \quad (1.1)$$

$$\beta_t = \beta_{t-1} + \eta_t \quad (1.2)$$

where P_{y_t} is the dependent variable, β_t is time-varying regression coefficient, and P_{x_t} is the independent variable at time t , respectively. Equation (1.1) can be seen as measurement equation, and equation (1.2) as state equation, which defines beta as a simple random walk in our study. ε_t and η_t are independent uncorrelated error terms with standard deviations H and Q , respectively. Here, the variances of the noise process and other unknown parameters have to be estimated. This is accomplished by maximizing the following likelihood function

$$\log L = -\frac{NT}{2} \log 2\pi - \frac{1}{2} \sum_{t=1}^T \log F_t - \frac{1}{2} \frac{v_t' v_t}{F_t} \quad (2)$$

Where v_t is the one-step ahead residual, i.e. difference between P_{y_t} and its estimate, and F_t is its variance. N and T are the number of columns of P_{x_t} (1 in our study) and the number of elements of the time series P_{y_t} . Those first two items are calculated recursively using the following equations.

$$\beta_{t|t-1} = \beta_t, v_t = P_{y_t} - P_{x_t} \beta_t \quad (3.1)$$

$$F_t = P_{x_t} P_{x_t}' + H \quad (3.2)$$

$$\beta_t = \beta_{t-1} + P_t P_{x_t}' \frac{v_t}{F_t} \quad (3.3)$$

$$P_t = P_{t-1} + P_t P_{x_t}' P_{x_t} P_t \frac{1}{F_t} + Q \quad (3.4)$$

Equations (3.1)-(3.4) are known as the Kalman filter. They are a series of regressions that generate an estimate of the coefficient β_t at time t given estimate of the coefficient at $t-1$. Once all these estimates have been obtained for our timeframe while having maximized the function (2), we can also generate an estimate of the updated residual series, i.e. spread between stocks, by using the following equation (4).

$$z_t = P_{Y_t} - \beta_t P_{X_t} \quad (4)$$

The spread is then standard normalized, directed to screening filters and tested to see if it can be used to generate trading signal. The standard normalized spread passes through two consecutive screening filters; one is the ADF unit-root test filter which evaluates mean-reversion property of the spread. In the case that the spread does not satisfy decision criteria, then it is flagged as invalid spread for trading signal generation and becomes ignored. Surviving spread is redirected to the next filter, mean-reversion strength test filter, where the spread is formulated into the OU process, equation (5), and mean-reversion strength κ is estimated.

$$dz_t = \kappa(\mu - z_t) + \sigma dW_t + \varepsilon_t \quad (5)$$

When the strength κ exceeds pre-determined threshold, which is determined based on calibration of the model, corresponding spread is set ready to generate trading signal. The graphical representation of the basic trading model is shown in Appendix B.

2.3. Trading Strategy

2.3.1. Basic Strategy

Our Basic trading strategy involves static threshold bounds which provide a guide for triggering a buy or sell decision when the surviving spread goes above or under the threshold bounds. In other words, market entry and exit signal are generated based on crossing between the threshold bounds and the spread. Basically, similar market timing strategy is applied to the traditional pairs trading strategy.

We sell (or buy) the spread when it is +2 above its mean (or when it is -2 below its mean)⁴, and the position is liquidated when the spread gets closer than 0.5 in magnitude to its mean. In other words, we sell (or buy) the spread when it break through +2 level downward (or -2 level upward), and the position is cleared when the spread gets closer than 0.5 to its mean. Figure 2 illustrates this strategy involving static threshold bounds. On top of that, we wait for 1 time point before we enter into the position, to be on the safe side and make sure that the strategy is viable in practice. For instance, in case of 30-minute data, after the condition for entry has been fulfilled, we wait for 30-minutes before we enter the position.

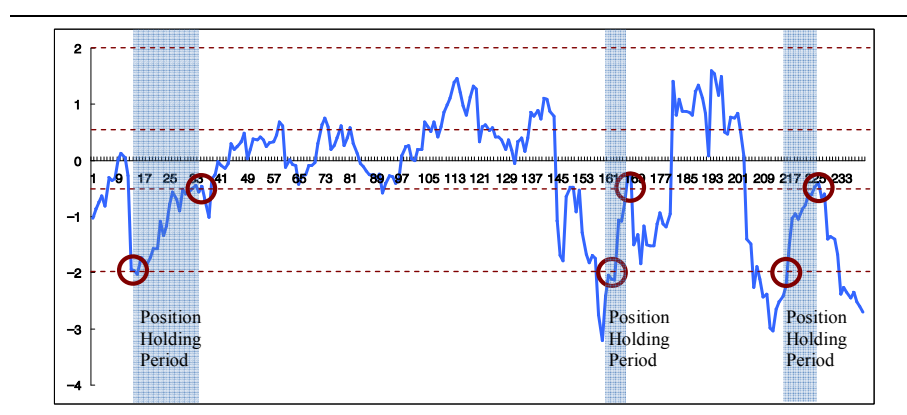


Figure 2. Basic trading strategy involving static threshold bounds

2.3.2. Loss-cut Strategy

Figure 3 below shows how winning ratio and average rate of return per trade of our strategy are evolving over position holding duration. It is observed that after 12 trading hours i.e. 2 trading days elapse, winning ratio and average rate of return per trade keep deteriorating sharply over time. And position previously set up is never cleared after this amount of time elapses, which raises the necessity to implement a customized loss-cut strategy to lock in profit. Our loss-cut strategy clears position if holding duration exceeds the pre-determined maximum trading duration. The maximum trading duration is set to be 12 trading hours from calibration which proves to optimally keep balance between winning ratio, average rate of return and average holding period.

⁴ Here +2 is equivalent to +2 standard deviation level, and mean level is zero as the spread is standard normalized.

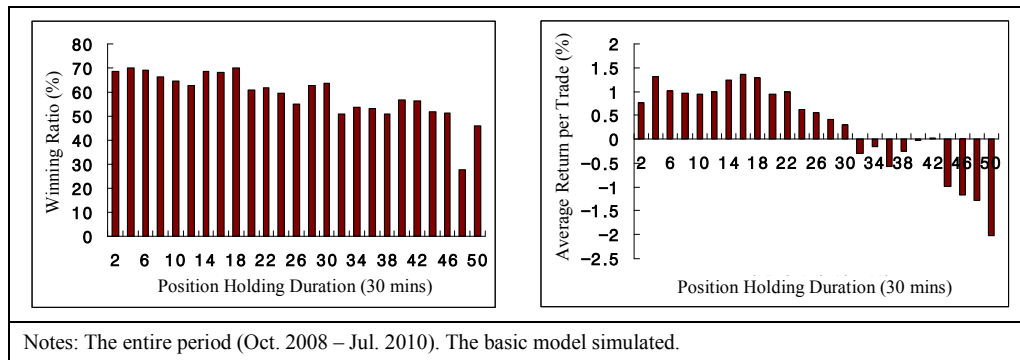


Figure 3. Development of winning ratio and average rate of return per trade over holding duration

2.4. The Enhanced Model

The enhanced model involves the same strategies for pairs selection, trading signal generation, and market timing. One difference between the basic and the enhanced model is that it selects high-ranking pairs to trade for the next time period based on a set of in-sample statistics which includes in-sample ADF test t-statistics, in-sample information ratio, and in-sample mean reversion strength. In our study, moving window with the size of 2 weeks are considered as in-sample period. Graphical representation of the enhanced model is shown in Appendix C.

IV. Results and Analysis

1. Performance Summary

In Table 1, we present the summary of performance statistics obtained from back-testing of the strategy⁵. The statistics include average holding period, average rate of return per trade, information ratio, and winning ratio etc. Average rate of return per trade is 0.68% and winning ratio is 63%. We have obtained 2.55% of average rate of return per trade in bearish period, which is higher than 0.80% in bullish period and 0.39% in flat period. In general, the direction of market movement is inversely correlated with market volatility, which suggests that our strategy is more preferable in the market with higher volatility. Meanwhile, Figure 4 below shows that the strategy succeeds in generating steady profit without being subject to market

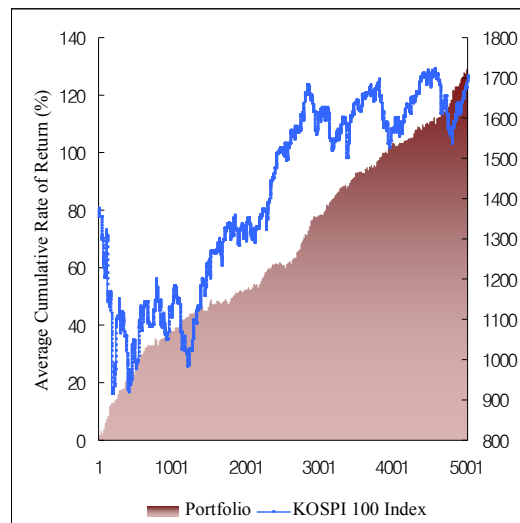
⁵ Back-testing details are summarized in Appendix A.

condition, thereby realizing market neutral strategy. More specifically, Figure 5 shows that the strategy achieves positive weekly excess return for most of the time periods. In particular, 4 weeks moving average of weekly excess return maintains positive level during the entire period.

	Bearish period	Bullish period	Flat period	Entire period
Average holding period (Hrs)	9.06	9.02	5.94	7.38
Average rate of return per trade (%)	2.55**	0.80**	0.39**	0.68**
Information ratio	5.79	2.17	2.76	2.45
Winning ratio (%)	73.74	63.96	61.92	63.39
The number of trades	495	3,224	3,889	7,616
Average number of trades a month	35	322	389	224
The number of success	365	2,062	2,408	4,828

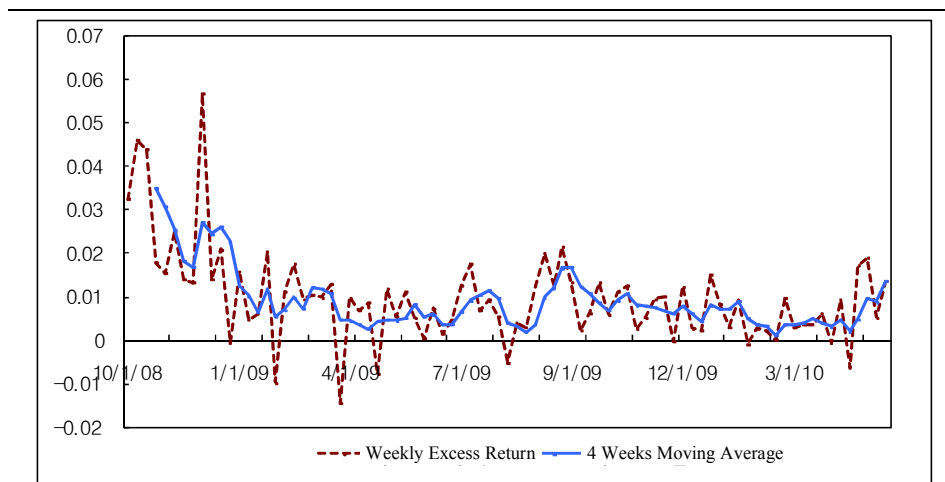
Notes: Bullish period (Dec. 2008 – Sep. 2009), Bearish period (Oct. 2008 – Nov. 2008), Flat period (Oct. 2009 – Jul. 2010). The entire period (Oct. 2008 – Jul. 2010). The basic model simulated. *, ** marked numbers represent values obtained with 5% and 1% significance level, respectively.

Table 1. The summary of performance statistics obtained from back-testing of the strategy



Notes: The entire period (Oct. 2008 – Jul. 2010). The basic model simulated.

Figure 4. Cumulative average rate of return during the entire period



Notes: The entire period (Oct. 2008 – Jul. 2010). The basic model simulated.

Figure 5. Weekly excess rate of return and its 4 weeks moving average during the entire period

2. Performance of the Strategy by Industry Group

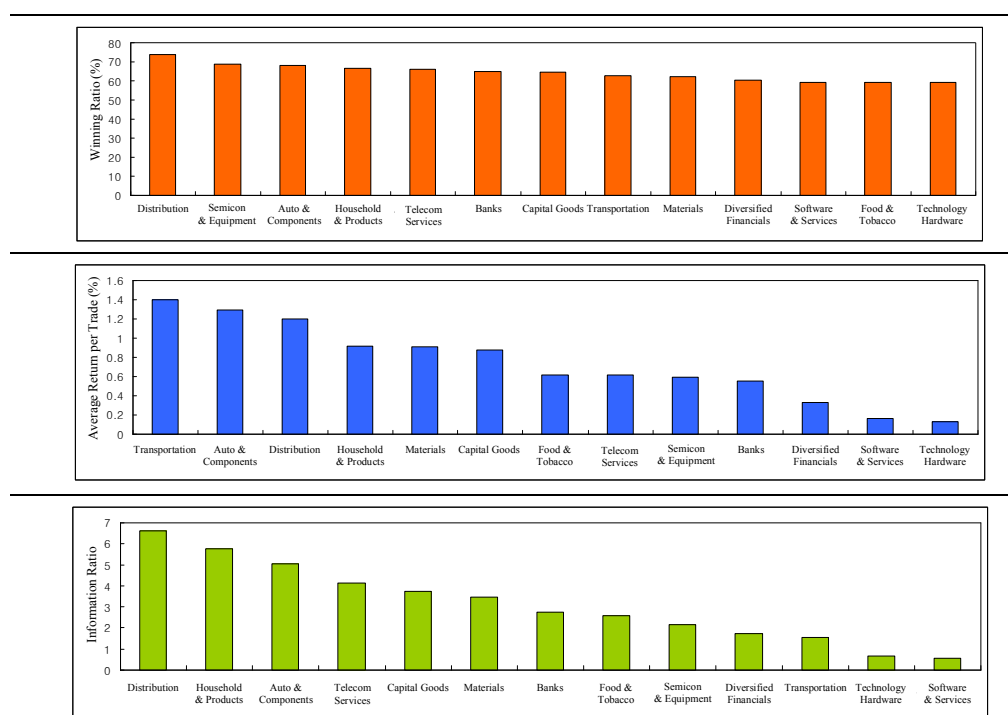
For trading signal generation, our strategy utilizes prices data only. Also it only focuses on pairs that are selected in the same industry group. Thus one could expect that performance of our strategy is differentiated between industry groups where each industry group has different dynamic drivers and runs through different paradigm shift at different timing. Table 2 summarizes performance statistics arranged by industry group, where we follow FIGCS (FnGuide Industry Group Classification Standard). Industry groups dominantly influenced by domestic demand i.e. non-cyclical in Korea such as Distribution, Household & Personal Products, and Automobiles & Components shows relatively higher winning ratio and average rate of return per trade whereas industry groups involving fast-paced technological development and variable international demand such as Technology Hardware and Software & Services give out relatively low statistics. It follows that stocks belong to these industry groups are less likely to maintain consistent cointegration relationship. Key performance statistics including winning ratio, average rate of return per trade, and information ratio are illustrated by industry group in Figure 6 below.

	Total Number of Pairs	Net Asset Value	Total Number of Trades	Winning Ratio (%)	Average Holding Period (Hrs)	Average Rate of Return (%)	Information Ratio	Cumulative Rate of Return (%)
Materials	14	40.51	1569	62.39	7.96	0.91**	3.46	189.19
Capital Goods	9	22.39	949	64.59	8.43	0.88**	3.75	148.46

Transportation	3	9.01	339	62.83	8.41	1.40	1.55	200.37
Automobiles & Components	1	3.79	113	68.14	6.83	1.29**	5.04	281.74
Distribution	3	10.78	328	73.78	6.71	1.19**	6.60	259.43
Food, Beverage & Tobacco	2	3.96	245	59.18	7.14	0.61*	2.59	98.02
Household & Personal Products	1	2.66	114	66.67	7.61	0.91**	5.75	165.86
Banks	14	29.45	1689	64.83	6.69	0.56**	2.76	110.48
Diversified Financials	10	15.24	1307	60.21	7.09	0.33**	1.74	52.37
Software & Services	1	1.70	128	59.38	6.85	0.16	0.56	70.44
Technology Hardware	3	3.49	426	59.15	6.65	0.13	0.67	16.62
Semiconductors & Equipment	1	1.65	100	69	7.82	0.59	2.17	64.79
Telecommunication Services	3	5.66	309	66.02	7.11	0.61**	4.15	88.57
Total	65	150.3	7616	63.39	7.38	0.68**	2.45	131.21

Notes: The entire period (Oct. 2008 – Jul. 2010). The basic model simulated. *, ** marked numbers represent values obtained with 5% and 1% significance level, respectively.

Table 2. The summary of performance statistics by industry group



Notes: The entire period (Oct. 2008 – Jul. 2010). The basic model simulated.

Figure 6. Comparison of key performance statistics by industry group

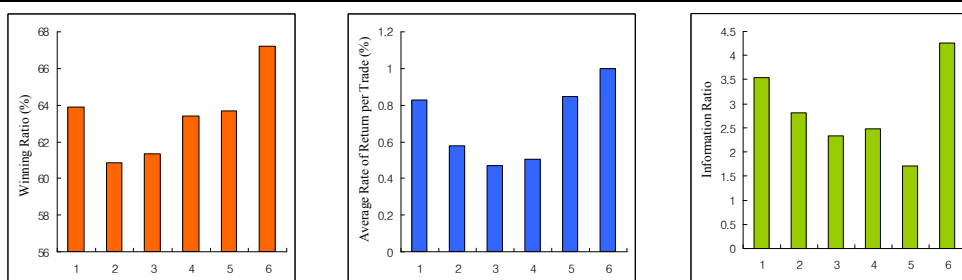
3. Intraday Performance

The use of high frequency equity data in modeling strategy allows us to analyze the performance of the strategy in high frequency domain. We first organize total deals traded during the day into groups by their market entry timing, and calculate performance statistics for each group. We summarize the statistics by groups in Table 3. Also key statistics including winning ratio, average rate of return, and information ratio are highlighted in Figure 7. Both Table 3 and Figure 7 indicate that the trades entered between 9:00AM and 10:00AM, around the market open, and between 14:00PM and 15:00PM, around the market close, show relatively higher winning ratio, average rate of return, and followed by higher information ratio. Recalling that intraday volatility pattern is generally formed in U-shape, the strategy seems to achieve higher performance in intraday time zone with higher volatility, which also corresponds to our previous finding that our strategy returns outstanding figures in volatile market trend. Besides, the strategy seems to take advantage of inefficiency derived from where stock price reflects the undisclosed market information. To be more specific, the trades entered at around the market close obtain the highest statistics. In general, overpriced stock tends to aggressively revert back to its fairly priced level as we run over to the end of the market. It implies that the trades set up around the market close take advantage of this trend and indeed do strongly revert to mean level with higher probability, producing superior performance statistics. Taken altogether, it would be advantageous for intraday trader or arbitrageur using high frequency strategy to focus on these intraday time zones for their trading.

Market Entry Time	Average Holding Period (Hrs)	Total Number of Trades	Total Winning Trades	Winning Ratio (%)	Average Rate of Return (%)	Information Ratio
9:00 ~ 10:00	7.42	1147	733	63.91	0.83**	3.55
10:00 ~ 11:00	7.06	1208	735	60.84	0.58**	2.81
11:00 ~ 12:00	6.91	1303	799	61.32	0.47**	2.34
12:00 ~ 13:00	7.22	1386	879	63.42	0.51**	2.48
13:00 ~ 14:00	8.12	1330	847	63.68	0.85**	1.71
14:00 ~ 15:00	8.24	1242	835	67.23	0.99**	4.26
Total	7.38	7616	4828	63.39	0.68**	2.45

Notes: The entire period (Oct. 2008 – Jul. 2010). The basic model simulated. *, ** marked numbers represent values obtained with 5% and 1% significance level, respectively.

Table 3. The summary of intraday performance statistics



Notes: The entire period (Oct. 2008 – Jul. 2010). The basic model simulated. 1: 09:00 – 10:00, 2: 10:00 – 11:00, 3: 11:00 – 12:00, 4: 12:00 – 13:00, 5: 13:00 – 14:00, 6: 14:00 – 15:00.

Figure 7. Comparison of key performance statistics by market entry timing

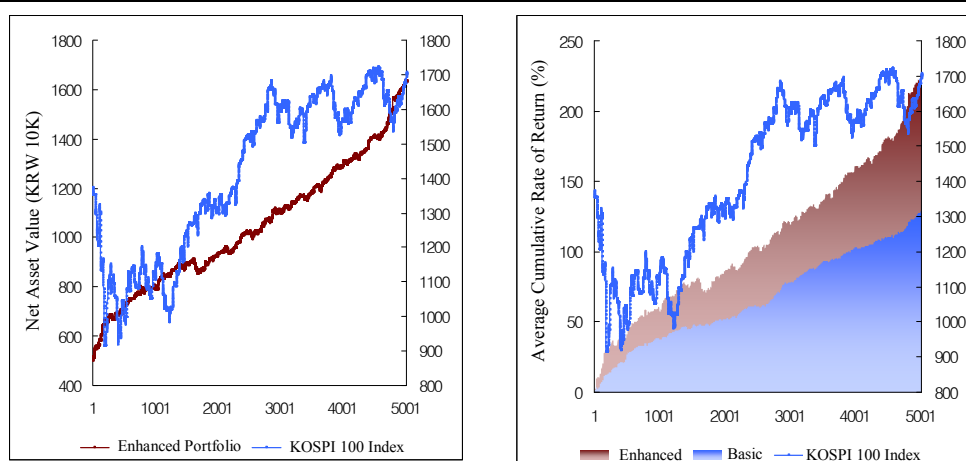
4. Performance of the Enhanced Model

As mentioned earlier, the enhanced model involves the same strategies for pairs selection, trading signal generation, and market timing as the basic model except that it selects high-ranking pairs to trade for the next time period based on a set of in-sample statistics, which include in-sample ADF test t-statistics, in-sample information ratio, and in-sample mean reversion strength. Performance comparison between the basic model and the enhanced model is illustrated in Figure 8. Cumulative average return of the enhanced model is higher than that of the basic model all the time during the entire testing period. Also Table 4 shows that other statistics such as average rate of return, information ratio, and winning ratio are improved as well.

	Total Number of Pairs	Net Asset Value	Total Number of Trades	Winning Ratio (%)	Average Holding Period (Hrs)	Average Rate of return (%)	Information Ratio	Cumulative Rate of return (%)
Basic Model	65	150.30	7616	63.39	7.39	0.68**	2.45	131.21
Enhanced Model	5	16.36	566	69.96	6.71	1.07**	3.79	227.05

Notes: The entire period (Oct. 2008 – Jul. 2010). The basic model simulated.

Table 4. Comparison of performance between the basic model and the enhanced model



Notes: The entire period (Oct. 2008 – Jul. 2010).

Figure 8. Performance of the enhanced model (left), performance comparison between the models (right)

V. Conclusion

This study is the first practice that modifies the well-known pairs trading strategy, one of typical market neutral strategies, to enable it to utilize high frequency equity data, and applies it to Korean stock market. This study is distinguished from the most of previous work on the traditional pairs trading strategy in that we generate trading signal by estimating the volatility of high frequency data using the Kalman filter and calculating time adaptive regression coefficient based on it. As for underlying universe for the strategy, we confine ourselves to consider the most liquid 100 stocks in KOSPI as a basket for our experiment. We analyze the results of out-of-sample performance test in various aspects: by market condition, industry group, and timing of the market entry. The results are quantified by utilizing a set of performance statistics such as average rate of return, winning ratio and the information ratio etc. Major findings suggest that arbitrage profitability is in fact present without being subject to market condition even when conservative transaction costs are taken into account. In particular, our strategy performs better in bearish market while it varies depending on industry groups. Industry groups relating to domestic consumption such as Distribution, Household & Personal Products, and Automobiles & Components show relatively higher performance, whereas industry groups involving fast-paced technological development and international demand such as Technology Hardware

and Software & Services give out relatively lower statistics. Also as for intraday performance, trades entered around the market open and close, show relatively outstanding figures. Furthermore, we introduce an enhanced version of the strategy for better profit potential and reliability, which selects 5 high-ranking pairs as our trading portfolio for the next time period, based on a set of in-sample statistics. It is verified that the enhanced strategy has better profitability and stability compared to our basic strategy.

References

- [1] Andrea Berardi, Stefano Corradin, and Cristina Sommacampagna (2002), “Estimating Value at Risk with the Kalman Filter”, Working paper.
- [2] Bentz, Y. (2003), “Quantitative Equity Investment Management with Time-Varying Factor Sensitivities”, *Applied Quantitative Methods for Trading and Investment*. John Wiley & Sons, Chichester, 213-237.
- [3] Binh Do and Robert Raff(2010), “Does Simple Pairs Trading Still Work?”, *Financial Analysts Journal*, Vol.66, No.4, pp.83-95.
- [4] Burgess, A. N. (1999), “A Computational Methodology for Modeling the Dynamics of Statistical Arbitrage”, London Business School, PhD Thesis.
- [5] Burgess, A. N. (2003), “Using Cointegration to Hedge and Trade International Equities”. In Dunis, C., Laws, J. And Naïm, P. [eds.] *Applied Quantitative Methods for Trading and Investment*. John Wiley & Sons, Chichester, pp. 41-69.
- [6] Boguen Choi (2006), “Arbitrage Strategy using Pairs Trading”, Seokang Univ., Master Thesis.
- [7] Choudhry, T. and Wu, H. (2009), “Forecasting the Weekly Time-Varying Beta of UK Firms: GARCH Models vs. Kalman Filter Method.”, *The European Journal of Finance*, Vol.15, No.4, pp.437-444.
- [8] Christian L. Dunis, Gianluigi Giorgioni, Jason Laws, and Jozef Rudy (2010), “Statistical Arbitrage and High-Frequency Data with an Application to Eurostoxx 50 Equities”, Liverpool Business School, Working paper.
- [9] Dunis, C. L. and Ho, R. (2005), “Cointegration Portfolios of European Equities for Index Tracking and Market Neutral Strategies”, *Journal of Asset Management*, 6, 1, 33-52.
- [10] Dunis, C. L. and Shannon, G. (2005), “Emerging Markets of South-East and Central Asia: Do They Still Offer a Diversification Benefit?”, *Journal of Asset Management*, 6, 3, 168-190.
- [11] Dunis, Giorgioni, Laws and Rudy (2010), “Statistical Arbitrage and High-Frequency Data with an Application to Eurostoxx 50 Equities”, Working paper.
- [12] Enders, W. and Granger, C. W. J.(1998), “Unit-Root Tests and Asymmetric Adjustment with an Example Using the Term Structure of Interest Rates”, *Journal of Business & Economic Statistics*, Vol.16, No.3, pp.304-311.

- [13] Engle, R. F. and Granger, C. W. J.(1987), “Co-integration and Error Correction: Representation, Estimation, and Testing”, *Econometrica*, Vol.55, No.2, pp.251-76.
- [14] Faust, J., Rogers, J. H., Wang, S.-Y. B. and Wright, J. H. (2007), “The High-Frequency Response of Exchange Rates and Interest rates to macroeconomic announcements”, *Journal of Monetary Economics*, Vol.5, No.4, pp.1051-1068.
- [15] Galenko, A., Popova, E. and Popova, I. (2007), “Trading in the Presence of Cointegration”, SSRN eLibrary, <http://ssrn.com/paper=1023791>.
- [16] Gatev E., Goetzmann W., and Rouwenhorst K. (2006), “Pairs Trading: Performance of a Relative-Value Arbitrage Rule”, *Review of Financial Studies*, Vol.19, No. 3, pp.797-827.
- [17] Ghysels, E., Santa-Clara, P. and Valkanov, R. (2006), “Predicting Volatility: Getting the Most Out of Return Data Sampled at Different Frequencies”, *Journal of Econometrics*, Vol.131, No.1-2, pp.59-95.
- [18] Gillespie, D. T.(1996), “Exact Numerical Simulation of the Ornstein-Uhlenbeck Process and Its Integral”, *Physical Review E*, Vol.54, No.2, pp.2084-91.
- [19] Gomez, G. N. (2009), “Beta and VaR Prediction for Stock Portfolios Using Kalman's Filter and Garch Models”, *Cuadernos de Administración*, Vol.18, No.29, pp.103-120.
- [20] Johansen, S. (1988), “Statistical Analysis of Cointegration Vectors. *Journal of Economic Dynamics and Control*”, Vol.12, pp.231-254.
- [21] Khandani, A. E. and Lo, A. W. (2007), “What Happened to the Quants in August 2007?”, *Journal of Investment Management*, Vol.5, No.4, pp.5-54.
- [22] Kwangchul Son (2010), “Performance of Pairs Trading in Korean Stock Market”, Kookmin Univ., Master Thesis.
- [23] Laviola, J. (2003a), “Double Exponential Smoothing: An Alternative to Kalman Filter-Based Predictive Tracking”, In *Proceedings of the Immersive Projection Technology and Virtual Environments*. ACM Press, 199-206.
- [24] Lin, Y.-X., McCrae, M. and Gulati, C.(2006), “Loss Protection in Pairs Trading through Minimum Profit Bounds: A Cointegration Approach”, *Journal of Applied Mathematics and Decision Sciences*, Vol.2006, pp.1-14.
- [25] Mudchanatongsuk, S., Primbs, J. A. and Wong, W. (2008), “Optimal Pairs Trading: A Stochastic Control Approach.”, In *Proceedings of the American Control Conference*, pp.1035-1039.

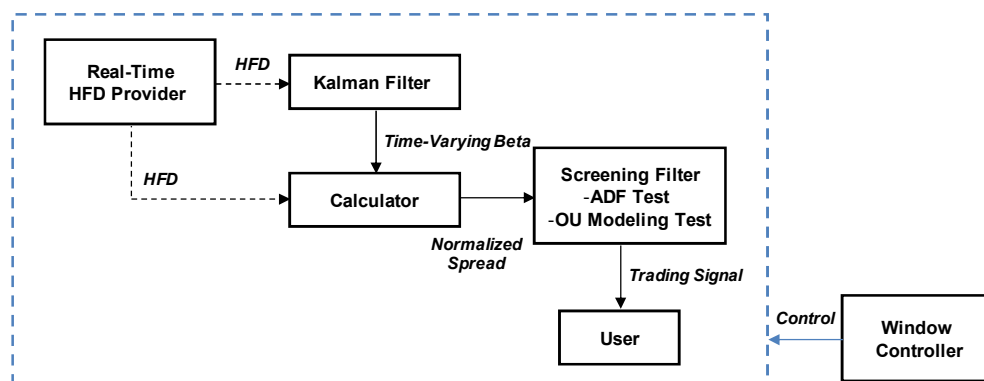
- [26] Nath, P. (2003), "High Frequency Pairs Trading with U.S. Treasury Securities: Risks and Rewards for Hedge Funds", Working paper.
- [27] Vidyamurthy, G. (2004), "Pairs Trading - Quantitative Methods and Analysis", John Wiley & Sons, Inc., New Jersey.
- [28] Wonhyuk Yoo (2007), "Performance Analysis of Pairs Trading: Relative Value Arbitrage Strategy", Seoul National Univ., Master Thesis.

Appendices

Appendix A. Back-testing Details

Item	Description	Note
Underlying Universe	Among 100 stocks comprising KOSPI 100 index, stocks having average trading amounts in the past 60 days less than 2 million dollars are discarded. Pairs in which each stock belongs to different industry group are discarded. Pairs having beta spread, calculated based on CAPM, larger than 0.2 are discarded	Among all the possible pairs from 100 stocks, we are left with 65 pairs at the final stage of the selection process.
Back-testing Data	Stock price data sampled at 30-mins interval within the horizon from the 1st of October 2008 to the 31st of July 2010	In order to properly adjust intraday data, first we obtain the ratio of daily closing price (adjusted by FnGuide) to the last intraday price for that day (representing the unadjusted closing price). Then we multiply all intraday data during that particular day by the calculated ratio. We repeat the procedure for all days and shares for which we have intraday data.
In-Sample and Out-of-Sample Period	Moving window with the size of one week i.e. 5 trading days is assumed to be the out-of-sample period and the period of the previous two weeks is assumed to be the in-sample period.	The in-sample period and the out-of-sample period are updated and assigned new data accordingly at every end of the week.
Transaction Costs	Conservative total transaction costs of 0.5% i.e. 50bps one-way in total for both shares. Total 100bps from market entry to position clearing.	Including taxes, brokerage fee, clearing cost, bid-ask spread.
Trading Strategy	Basic market entry strategy involving static threshold bounds and customized loss-cut strategy are implemented.	The loss-cut strategy clears position if holding duration exceeds the pre-determined maximum trading duration, 12 trading hours, to optimally keep balance between winning ratio, average rate of return and average holding period.

Appendix B. The Graphical Representation of the Basic Model



Appendix C. The Graphical Representation of the Enhanced Model

